Wednesday, September 6, 2023 7:33 PM

3. Software estimation metrics 3Q | 20-25m

Analytical chp.

① Software Project Estimation

- LOC based estimation
- FP based estimation || 1Q | 10m
- COCOMO model

→ Effort
→ Schedule/Durn
→ Resources
→ Cost
→ Man pow/ Human

② Project Scheduling and Tracking → 1Q | 10m → 1Q | 6m.

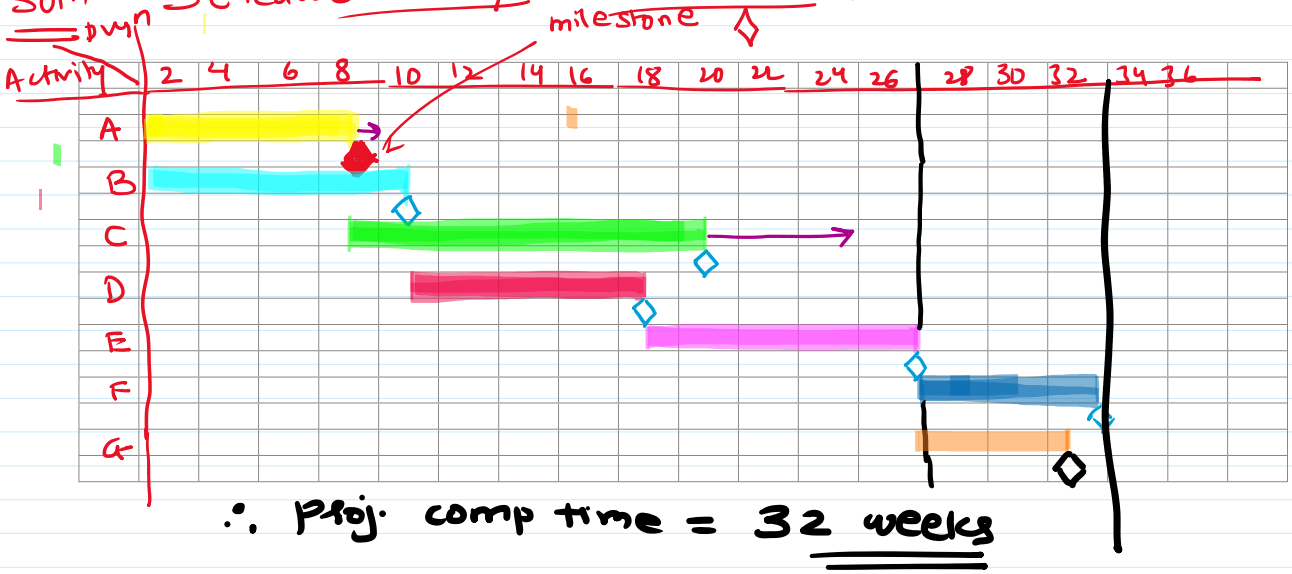
- Principles of Scheduling + tracking
- Scheduling Tech
  - Timeline chart
  - CPM
  - PERT
- → Explain the procedure of project tracking

* Consider the following project XYZ

Activity	A	B	C	D	E	F	G
Duration (weeks)	7	9	12	8	9	6	5
Interdep.	—	—	A	A,B	D	C,E	E

* Find project comp. time. Draw timeline chart.

Soln ⇒ Schedule chart / Timeline chart / Gantt chart



∴ Proj. comp time = 32 weeks

* Principles of project Scheduling Imp Q

① Compartmentalization
② Interdependency

CITED³

1. Compartmentalization
2. Interdependency
3. Time allocation
4. Effort validation
5. Defined responsibilities
6. Defined outcomes
7. Defined milestones

CITED3

* CPM – Critical Path Method

1. Perform CPM on proj XYZ
2. Draw AON / Network diag
3. Find critical path in n/w
4. Find proj comp. time

* Find project comp. time.
Draw timeline chart.

* Consider the following project XYZ

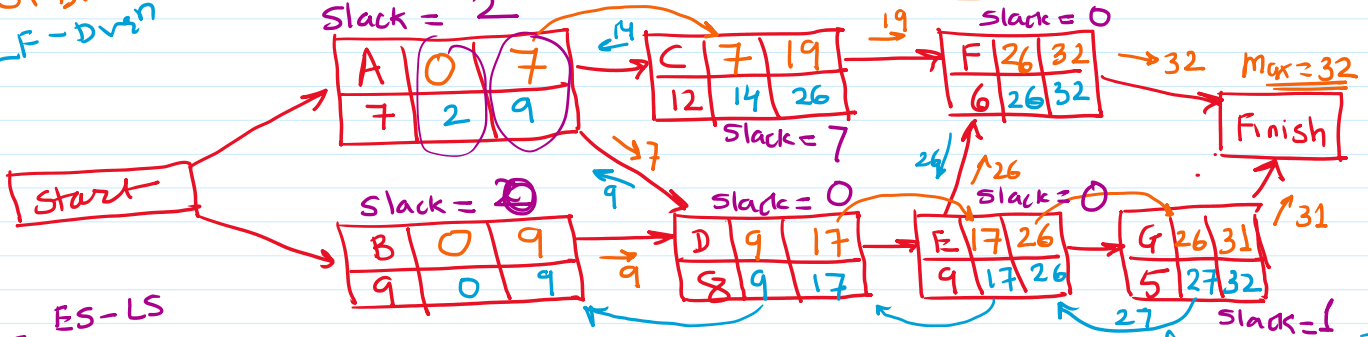
Activity	A	B	C	D	E	F	G
Duration (weeks)	7	9	12	8	9	6	5
Interdep.	—	—	A	A,B	D	C,E	E

Soln Draw AON [Activity on Node] / Network diag.

Act | ES | EF
Dur | LS | LF

EF = ES + Dur
LS = LF − Dur

Forward pass [ES, EF – max]



Backward pass [LS, LF; min]

Slack time = ES − LS = EF − LF

Critical Activity ⇒ Activity with slack = 0 [B, D, E, F]

Non-critical Activity ⇒ Activity with slack ≠ 0 [A, C, G]

Critical path ⇒ Start → B → D → E → F → Finish

proj comp time ⇒ 9 + 8 + 9 + 6 = 32 weeks

* Consider the following project-

Activity	A	B	C	D	E	F	G
time (week)	7	9	12	8	9	6	5
Interdep.	—	—	A	A,B	D	C,E	F

EF = ES + D

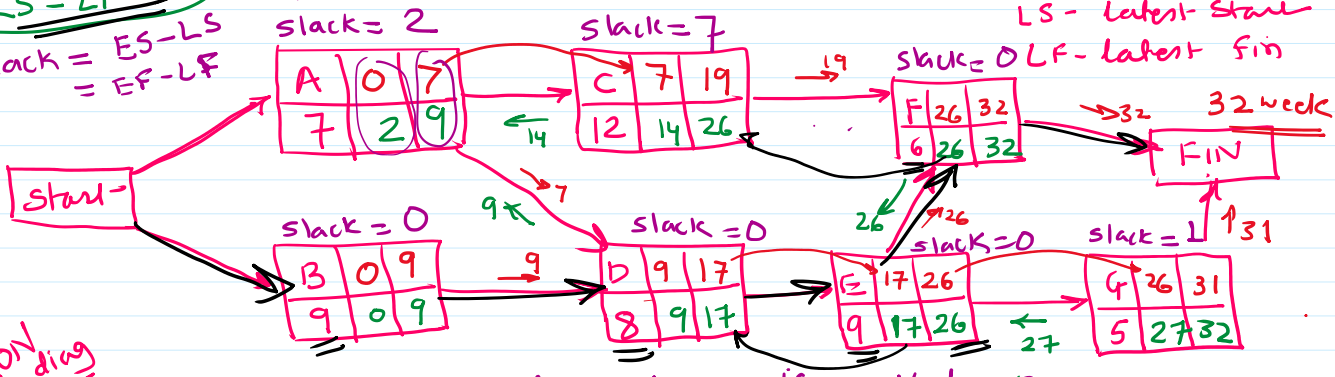
LS = LF - D

Slack = ES - LS = EF - LF

Activity → flw pass

Activity	ES	EF
Dur n	LS	LF

Forward pass (max) ES - Early start, EF - Early finish, LS - Latest start, LF - Latest fin, Backward pass (min)



AON diag

CRITICAL Activity ⇒ Any activity is called as critical activity iff slack <= 0 for that activity

Critical path = start → B → D → E → F → Fin

⇒ 9 + 8 + 9 + 6 ⇒ 32 week

* PERT - Project Evaluation & Review Tech [R&D projects, proj where est. time reqd to comp. the activity is not known]

* The follo table shows the job of nlw along with time estimates.

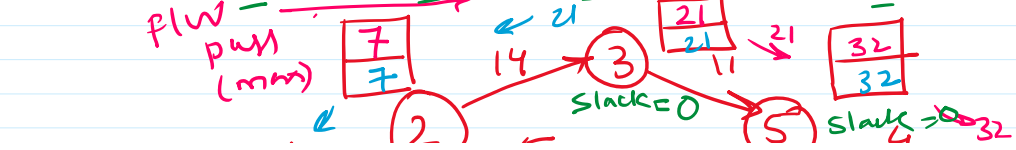
Activity		1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
time (weeks)	to	1	2	2	2	7	5	5	3	8
	tm	7	5	14	5	10	5	8	3	7
	tp	13	14	16	8	19	17	29	9	32

① Draw proj nlw diag
② Find te for each activity
③ Find proj comp time

Soln Estimated time $te = \frac{to + 4tm + tp}{6}$

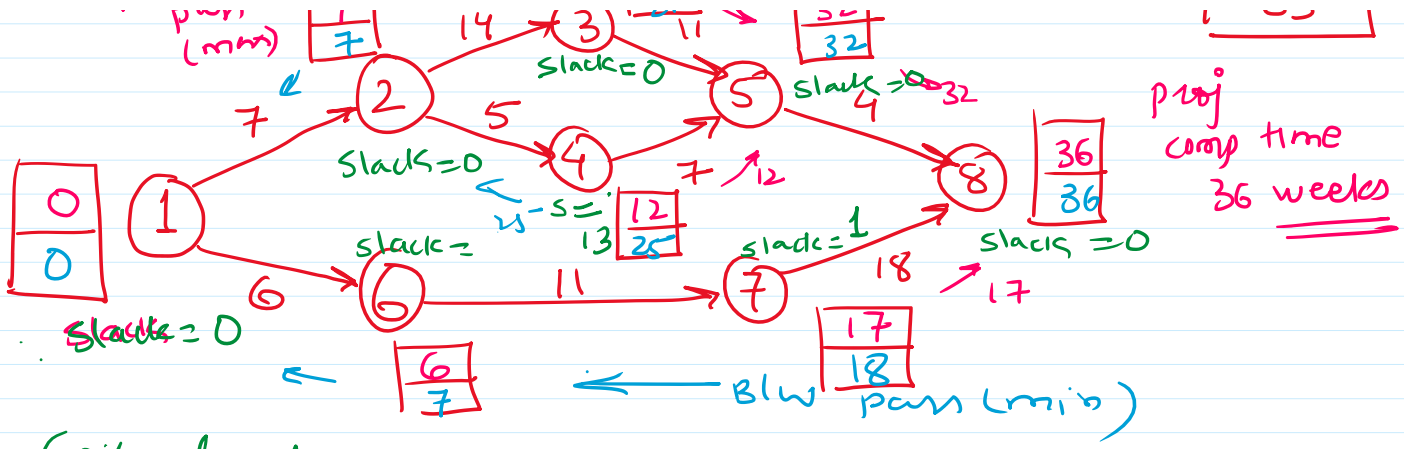
Activity	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
te	7	6	14	5	11	7	11	4	18

flw pass (max)



ES
LS

proj



Critical path ⇒ 1 → 2 → 3 → 5 → 8

proj comp time = 7 + 14 + 11 + 4 ⇒ 36 weeks

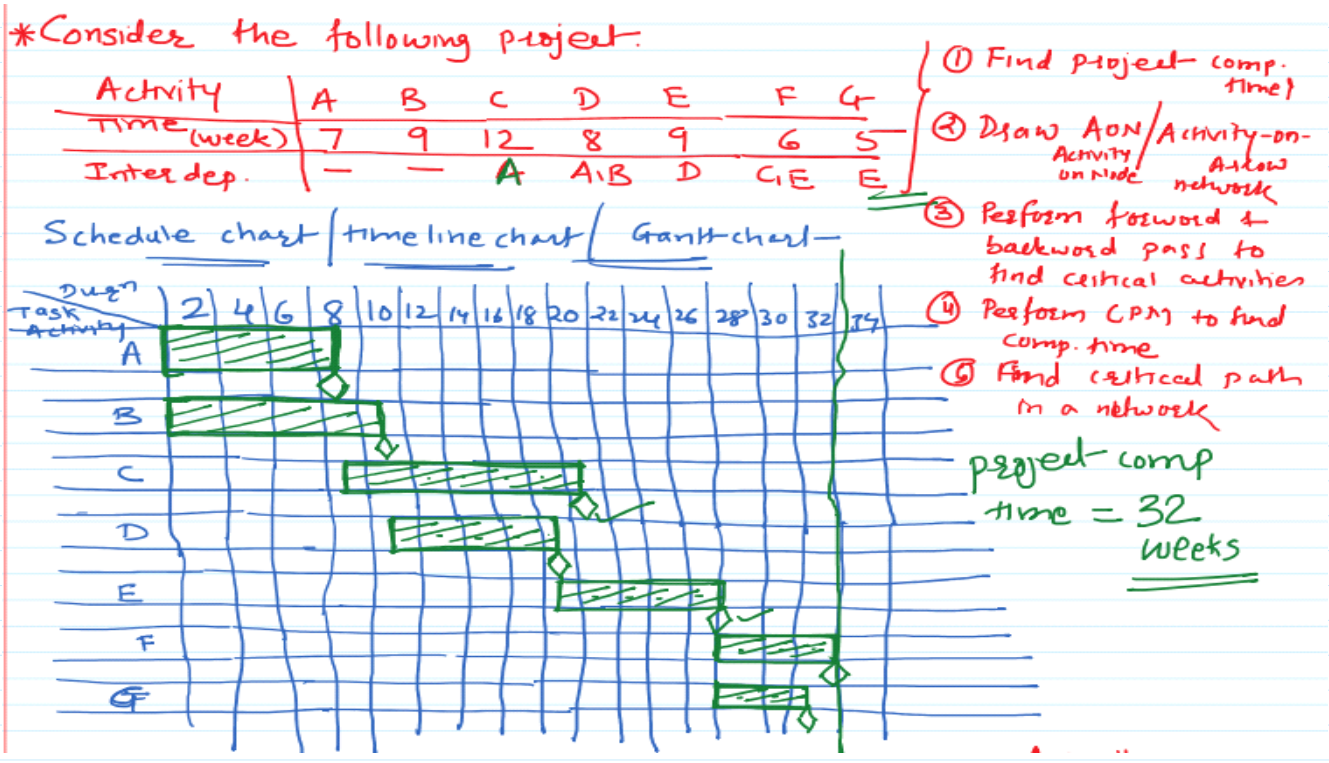
Q. Explain the basic principles of project scheduling. → CITEDDD

1. **Compartmentalization**: The project must be compartmentalized into a number of manageable activities and tasks.
2. **Interdependency**: The interdependency of each compartmentalized activity or task must be determined.
3. **Time allocation**: Each task to be scheduled must be allocated some number of work units.
4. **Effort validation**: Every project has a defined number of staff members.
5. **Defined responsibilities**: Every task that is scheduled should be assigned to a specific team member.
6. **Defined outcomes**: Every task that is scheduled should have a defined outcome.
7. **Defined milestones**: Every task or group of tasks should be associated with a project milestone. A milestone is accomplished when one or more work products has been reviewed for quality.

Q10. List various project scheduling methods (scheduling tools and techniques)

1. Time-line chart or Gantt chart
2. CPM (Critical Path Method)
3. PERT (Program Evaluation & Review Technique)

Time-line chart or Gantt chart



CPM (Critical Path Method)

CPM – Critical Path Method –

Activity	A	B	C	D	E	F	G
Dur^{n} (week)	7	9	12	8	9	6	5
Intredep	—	—	A	A,B	D	C,E	E

ES – Early start
EF – Early finish
LS – latest start
LF – latest finish

Activity

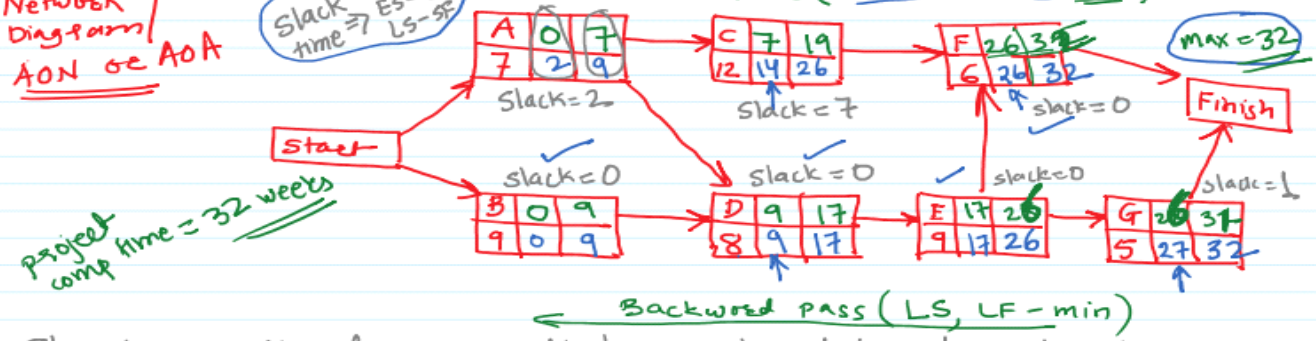
Act^{n}	ES	EF
Dur^{n}	LS	LF

LS = LF – Dur^{n}
EF = ES + Dur^{n}

Network Diagram / AON or AOA

Slack time ⇒ ES–EF, LS–SF

Forward pass (Find ES & EF – max)



Backward pass (LS, LF – min)

project comp time = 32 weeks

Slack time ⇒ How long an activity can be delayed without extending the project comp. time

Critical path ⇒ ~~Act^{n}~~ Activity path whose slack = 0

Critical path ⇒ B – D – E – F ← (These activities are critical acti)

project comp time ⇒ 9 + 8 + 9 + 6 ⇒ 32 weeks

PERT (Program Evaluation & Review Technique)

The follo table shows the job of network along with their time estimates.

Activity	Estimated dur^{n} in weeks			Estimated time (te)	Var $\sigma^2 = \left(\frac{tp - to}{6}\right)^2$
	opt time (to)	most likely (tm)	pessi time (tp)	$te = \frac{to + 4tm + tp}{6}$	
1 – 2	1	7	13	7	4
1 – 6	2	5	14	6	4
2 – 3	2	14	26	14	16
2 – 4	2	5	8	5	1
3 – 5	7	10	19	11	4
4 – 5	5	5	17	7	4
6 – 7	5	8	29	11	16
5 – 8	3	3	9	4	1
7 – 8	8	17	32	18	16

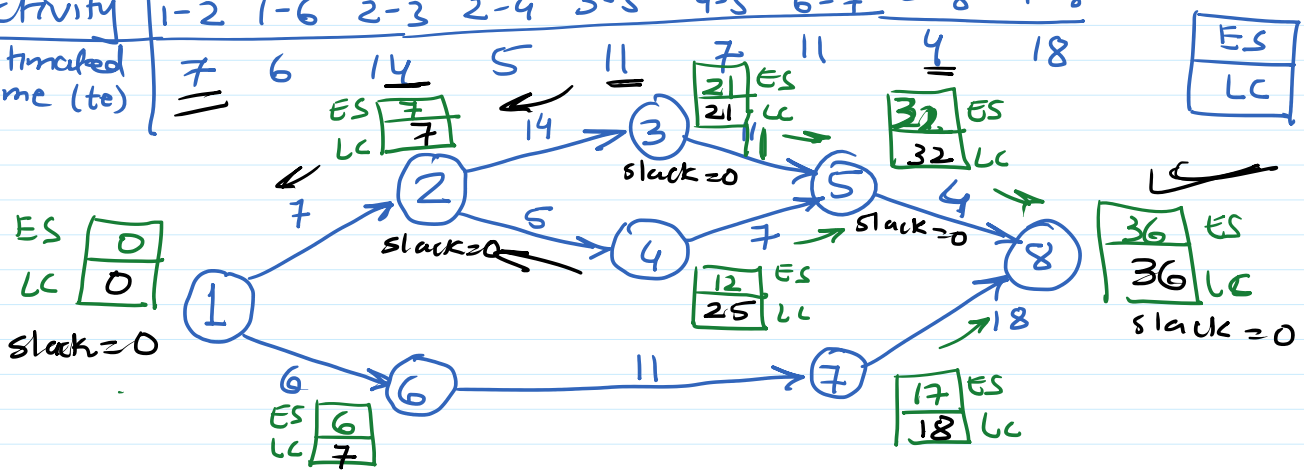
Ques ① Draw project network diag ✓

② Find expected dur^{n} reqd to comp. project ⇒ 36 weeks

③ Cal earliest + latest occ. of each activity
④ Cal expected project length
⑤ Cal variance & SD of project

$SD = \sqrt{var}$ $var = \sigma^2$

Activity	1-2	1-6	2-3	2-4	3-5	4-5	6-7	5-8	7-8
Estimated time (te)	7	6	14	5	11	7	11	4	18



Critical path ⇒ 1-2-3-5-8

proj comp time ⇒ 7+14+11+4 ⇒ 36 weeks

project length = σ^2 = 4+16+4+1 = 25

SD = σ = $\sqrt{25}$ = 5

Differentiate between CPM and PERT.

CPM differs from PERT as follow:

- The Program Evaluation and Review Technique (PERT) is a project management technique or tool which is suitable for projects that have unpredictable activities while the Critical Path Method (CPM) is a project management tool which is suitable for projects that have predictable activities.
- CPM uses a single estimate for the time that a project can be completed while PERT uses three estimates for the time that it can be completed.
- CPM is a deterministic project management tool while PERT is a probabilistic project management tool.
- CPM allows project management planners to determine which aspect of the project to sacrifice when a trade-off is needed in order to complete the project while PERT does not.

Parameters	PERT	CPM
Meaning	Program Evaluation & Review Technique used to plan & control large projects.	Critical Path Method is a project management tech. which plans & control the complex but routine projects.
Orientation	Event oriented	Activity oriented
Activities	Probabilistic	Deterministic
Focus	Time	Time & cost
Estimates	Three times (optimistic, most likely & pessimistic)	One time estimates (actual)
Suitable for	High precision time estimates	Reasonable time estimates
Nature	Non-repeatable activities	Repeatable activities
Usage	Best for R&D projects because activities are unpredictable in nature	Construction projects, routine projects, predictable activities

Explain the process of project tracking with suitable example.

The project schedule provides a road map for a software project manager. If it has been properly developed, the project schedule defines the tasks and milestones that must be tracked and controlled as the project proceeds. Tracking can be accomplished in a number of different ways:

- Conducting periodic project status meetings in which each team member reports progress and problems.
- Evaluating the results of all reviews conducted throughout the software engineering process.
- Determining whether formal project milestones have been accomplished by the scheduled date.

- Conducting periodic project status meetings in which each team member reports progress and problems.
- Evaluating the results of all reviews conducted throughout the software engineering process.
- Determining whether formal project milestones have been accomplished by the scheduled date.
- Comparing actual start-date to planned start-date for each project task listed in the resource table
- Meeting informally with practitioners to obtain their subjective assessment of progress to date and problems on the horizon.
- Using earned value analysis assess progress quantitatively.

Project Tracking chart / Table –

Activity / work task	Planned start	Actual start	Planned completion	Actual completion	Assigned	Effort
A	WK1, D1	WK1, D1	WK4, D2	WK4, D2	ABC	100%
B	WK1, D1	WK1, D1	WK5, D2	WK5, D4	PQR	95%
C						
D						

In reality, all of these tracking techniques are used by experienced project managers. Control is employed by a software project manager to administer project resources, cope with problems, and direct project staff. If things are going well (i.e., the project is on schedule and within budget, reviews indicate that real progress is being made and milestones are being reached), control is light. But when problems occur, the project manager must exercise control to reconcile them as quickly as possible. After a problem has been diagnosed, additional resources may be focused on the problem area: staff may be redeployed, or the project schedule can be redefined.

Explain how Task Set is define and evaluated in software projects. (TSS) –

* Explain TSS [Task Set Selector] with suitable ex. OR
OR
* Carry out project scheduling for the analysis ~~of~~ phase of a project involving database creation of a company with a strong workforce of 100 emp.

Soln → steps in project scheduling –

Step 1. Defining a task set –
Includes Identify manageable tasks, dwn reqd to comp the task, interdep, workproduct / outcome and milestones of each task
ex – Taskset for simple database project
1) Collect data
1.1 – Meet DBA (Database admin)
1.2 – Analyze the specifications given
1.3 – Prepare documentation for the analysis
1.4 – Get the document certified.
2) Make final submission
2.1 – Screen the document –
2.2 – Make final submission and update the result

Step 2 – Identify process model and the prototype
→ New appln generation
→ Appln enhancement –
→ Re-engineering
→ Project maint
} Consider db project as a new appln generation project

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Step 3- Determine the deg of rigor -
Adaptation criteria is set to define the complexity of a task

(a) Size of project-
(b) No. of potential user
(c) Mission criticality
(d) Appln longevity
(e) Stability of req.
(f) Ease of cust-dev comm
(g) Maturity of applicable tech
(h) Performance constraints
(i) Embedded & non emb char
(j) Staffing
(k) Re-engg factors

Needs to be considered during project planning and scheduling.

TSS
→ TSS ≤ 1.5 — deg of rigor is casual
→ 1.5 < TSS < 3 deg of rigor is structured
→ TSS > 3 deg of rigor is strict

Step 4- Design time line chart [Schedule chart]
Step 5- Tracking chart-

Adaptation Criteria	1-5 * Grade	0.8-1.2 * Weight	* Ndev	Entry pt multipliers Enhance	maint	Reengg	Total G * W * N
Proj size	4	1	1	1	1	1	4
potential user	4	1	1	1	1	1	4
Mission criticality	2	1	1	1	1	1	2
Appln longevity	3	1	1	1	0	0	3
Stability of req.	4	1	1	1	1	1	4
C-D comm	4	1	1	1	1	1	4
maturity of appln tech	2	1	1	0	0	1	2
performance constr	2	1	1	1	1	1	2
Emb & Non Emb char	1	1	1	1	1	1	1
staffing	2	1	1	1	1	1	2
Re-engg factors	2	1	0	0	0	1	0
						Total	28 / 11 ⇒ 2.54

TSS ⇒ 2.54

Deg of rigor of our TSS is Structured (Avg)

TSS < 1.5 — casual
1.5 < TSS < 3 — structured
TSS > 3 — strict

Q7. Enlist the reason for failure of software project

Although there are many reasons why software is delivered late, most can be traced to one or more of the following root causes:

1. An unrealistic deadline established by someone outside the software development group and forced on managers and practitioners within the group.
2. Changing customer requirements that are not reflected in schedule changes.
3. An honest underestimate of the amount of effort and/or the number of resources that will be required to do the job.
4. Predictable and/or unpredictable risks that were not considered when the project commenced.

group.

2. Changing customer requirements that are not reflected in schedule changes.
3. An honest underestimate of the amount of effort and/or the number of resources that will be required to do the job.
4. Predictable and/or unpredictable risks that were not considered when the project commenced.
5. Technical difficulties that could not have been foreseen in advance.
6. Human difficulties that could not have been foreseen in advance.
7. Miscommunication among project staff that results in delays.
8. A failure by project management to recognize that the project is falling behind schedule and a lack of action to correct the problem

OR

- Insufficient front-end planning
- Unrealistic project plan
- Project scope underestimated
- Customer/management changes
- Insufficient contingency planning
- Inability to track progress
- Inability to track problem early
- Insufficient number of checkpoints
- Staffing problems
- Technical complexity
- Sinking team spirit
- Unqualified resources
- Product is unknown to customer
- Requirements late
- Requirements of insufficient quality
- Delivered software late.
- Delivered software of insufficient quality

* Explain software project estimation tech OR s/w project measurement tech.

Parameters for s/w estimation
- SLOC or LOC
- Effort
- Duration
- No. of peoples
- Approx cost

1. Size-oriented metrics estimation
2. LOC/SLOC based estimation

Old tech / Historical or past prev proj info

3. FP (Function Point) based estimation ***
4. COCOMO estimation – Constructive cost model. *** 10

(1) Size-oriented metrics estimation

Project | LOC | $ or Rs | doc per KLOC | Errors | Defects | peoples

(2) Size-oriented metrics estimation

project	LOC	effort	$ or Rs cost	doc per KLOC Pp doc	Errors	Defects	peoples
—	(—)	—	—	—	—	—	—

metrics | indicator

(3) Function Point (FP) based estimation :- Imp

It computes FP for the project → Direct measurement
→ Indirect measurement

$$FP = \text{count total} * [0.65 + 0.01 * \Sigma Fi]$$

count total → direct-para
ΣFi → indirect-para

(a) Direct Measurement parameter :-

SEI

Measurement parameter	Count	Weighting factor: Simple	Average	Complex	count total
1) Number of user inputs	a	3	4	6	4a
2) Number of user outputs	b	4	5	7	5b
3) Number of user inquiries	c	3	4	6	4c
4) Number of files	d	7	10	15	10d
5) Number of external interfaces	e	5	7	10	7e
Count total					

(b) Indirect measurement parameter :-

ΣFi (1-5)

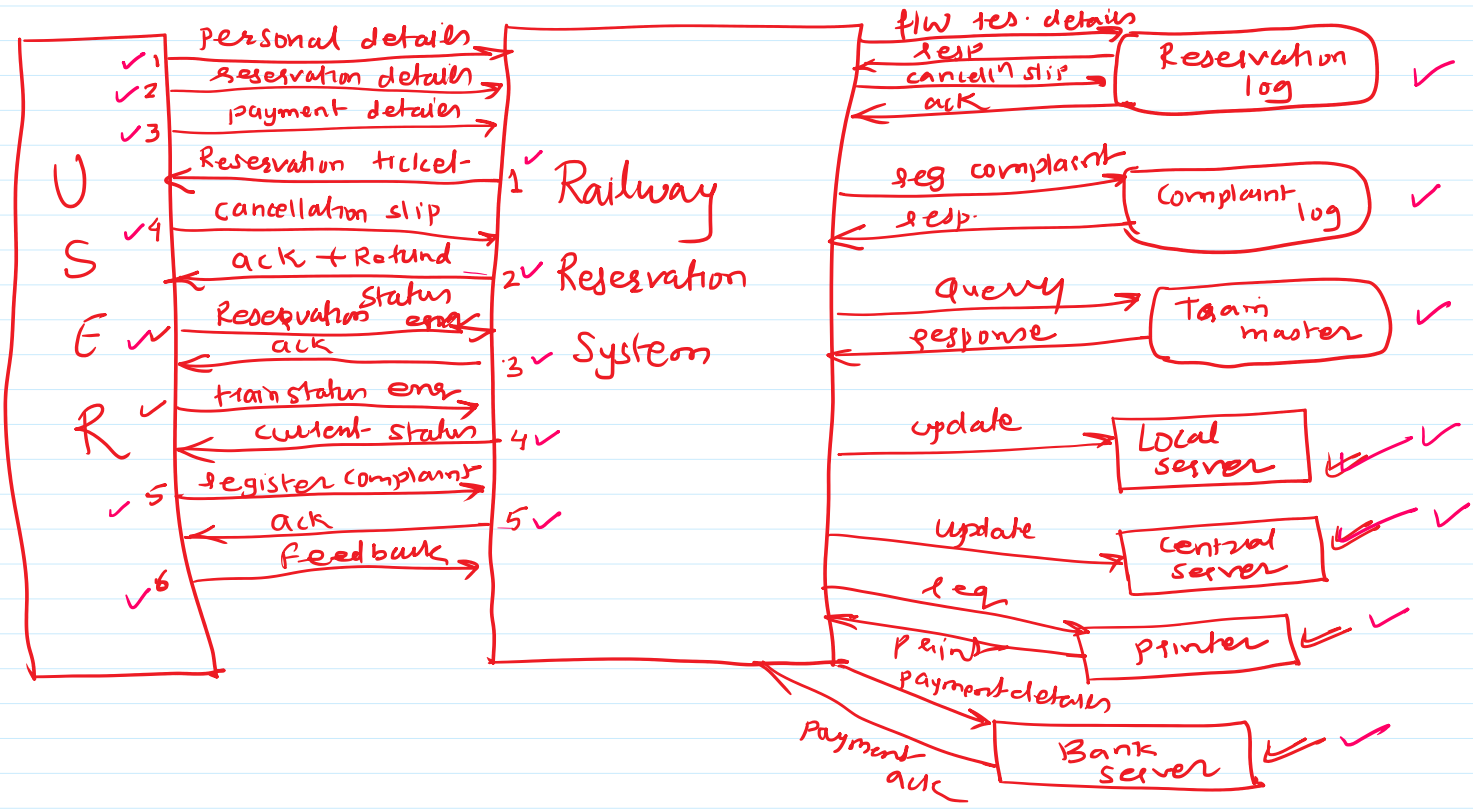
The Fi are "complexity adjustment values" based on responses to the following questions-

1. Does the system require reliable backup and recovery? → 5
2. Are data communications required? → 3
3. Are there distributed processing functions? → 4
4. Is performance critical? → 3
5. Will the system run in an existing, heavily utilized operational environment? → 3 4
6. Does the system require on-line data entry? →
7. Does the on-line data entry require the input transaction to be built over multiple screens or operations? — 3 4
8. Are the master files updated on-line? —
9. Are the inputs, outputs, files, or inquiries complex? — 3
10. Is the internal processing complex? — 3
11. Is the code designed to be reusable? — 3
12. Are conversion and installation included in the design? — 4
13. Is the system designed for multiple installations in different organizations? — 4
14. Is the application designed to facilitate change and ease of use by the user? — 3

ΣFi

* Compute Function Point for Railway Reservation System →

* Compute Function Point for Railway Reservation System →



Measurement parameter	Count		Weighting factor: Simple	Average	Complex		
Number of user inputs	6	×	3	4	6	=	24
Number of user outputs	5	×	4	5	7	=	25
Number of user inquiries	2	×	3	4	6	=	8
Number of files	3	×	7	10	15	=	30
Number of external interfaces	4	×	5	7	10	=	28
Count total →							115

Assume $\sum f_i = 45$

$FP = \text{Count total} * [0.65 + 0.01 * \sum f_i]$

$= 115 * [0.65 + 0.01 * 45]$

$= 115 * 1.10$

$FP = 126.5$

* COCOMO Model :- Constructive cost model

① Basic COCOMO :- It computes the s/w development efforts and cost as a function of program size expressed in line of code (LOC/KLOC)

expressed in line of codes (LOC | SLOC)

	Software project	A_b	B_b	C_b	D_b
simple	Organic	2.4	1.05	2.5	0.38
Avg *	Semi-detached	3.0	1.12	2.5	0.35
complex	Embedded	3.5	1.2	2.5	0.32

SEI

Intermediate COCOMO :- It also computes development effort as a function of prog size in KLOC

Software projects	a_i	b_i
Organic	3.2	1.05
* Semi-detached	3.0	1.12
Embedded	2.8	1.2

SEI

① Effort eqn $E = a_i (KLOC)^{b_i} * EAF$ — Effort adjustment (0.8 - 1.2) factor — person-months

② Duration $D = C_b (E)^{D_b}$ months

③ No. of people $N = \frac{E}{D}$

④ Cost- $Cost = E * \text{approx cost per person}$ [It does not include sal of emp]

In exam

LOC/SLOC ⇒ given/assume
a_i, b_i, C_b, d_b ⇒ given

EAF - given/Assume (0.8-1.2)

* Perform cocomo cost estimation for developing slw for billing customers of internet- service provider (Assume LOC = 7K, $a_i = 3.0$, $b_i = 1.12$, $C_b = 2.5$ and $d_b = 0.35$).

Soln SLOC/LOC = 7000
KLOC = 7

① Effort eqn $E = a_i (KLOC)^{b_i} * EAF$
$= 3.0(7)^{1.12} * 1$
$E = 26.52$ person-month

③ No. of person $= N = \frac{E}{D}$
$N = \frac{26.52}{7.87}$
$N = 3.36 \approx 4$ person

④ cost- Estimated

$E = 26.52$ person-month

(2) Duration $D = C_b (E)^{d_b}$

$= 2.5 (26.52)^{0.35}$

$D = 7.87$ months ✓

(4) Cost-

Estimated cost = E * cost per person

$= 26.52 * 2500$ (Assumed val)

Cost = 66300/- Rs

* Consider a project to develope a full screen editor. The major components are screen edit, command lang interpreter, file i/o, cursor ~~move~~ movement and screen movement. The size of these are estimated as 4K, 2K, 1K, 2K and 3K delivered source lines. Use COCOMO to det- overall effort, cost and dura reqd to comp. project

Total LOC = Screen edit + comm lang int- + file i/o + Cursor move + Screen move

$= 4 + 2 + 1 + 2 + 3 = 12$ KLOC

$E = a_i (KLOC)^{b_i} * EAF = 3.0(12)^{1.12} * 1 \Rightarrow$

$D = C_b (E)^{d_b} = 2.5(\quad)^{0.35} \Rightarrow$

$N = \frac{E}{D} =$